

UNIT-I

Cloud Computing Fundamentals: Definition Of Cloud Computing: Defining a cloud, Evolution Of Cloud Computing cloud types•NIST model, cloud cube model, Deployment models, Service models, Cloud Reference model. Characteristics of Cloud, Cloud Computing Benefits and Limitations, Cloud Architecture: Introduction on Infrastructure, platforms, virtual appliances, communication protocols; Cloud computing vs. Cluster computing vs. Grid computing; Applications: Technologies and Process required when deploying Web services; Deploying a web service from inside and Outside of a Cloud. Services and Applications by Types: IaaS, PaaS, SaaS. IDaaS, and CaaS.

UNIT II

Virtualization: Objectives, Benefits of Virtualization, Emulation, Virtualization for Enterprise, VMware, Server Virtualization, Data Storage Virtualization, balancing and Virtualization, Improving Performance through Load Balancing, Hypervisors, Machine Imaging, Porting Of applications in the cloud. Concept Of Software-Defined Networking (SDN), Network-Function Virtualization (NFV) and Virtual Network Functions (VNF).

Use of Platforms in Cloud Computing: Concepts of Platform as a Service, Use of PaaS application frameworks; Use of Google. Amazon and Microsoft Web Services. Cloud vendors and Service Management: Amazon cloud, AWS Overview, Installation of AWS, Google app engine, Azure cloud. Salesforce.

UNIT - III

Cloud Management: Features Of Network management system, Monitoring Of an entire cloud computing deployment stack, lifecycle management Of cloud services(six Stages Of lifecycle) Service Management in Cloud Computing: Service Oriented Architecture: concepts of message-based transactions, Protocol stack for an SOA architecture, Event driven SOA, Enterprise Service bus, Service Catalogs. Service Level Agreements (SLAs), Managing Data. Looking at Data, Scalability & Cloud Services, Database & Data Stores in Cloud , Large Scale Data processing.

UNIT - IV

Cloud Security Concepts: Cloud security challenges. Cloud security approaches: encryption. tokenization/ Obfuscation, cloud security alliance standards, cloud security models and related patterns, Cloud security in mainstream vendor solutions, Mainstream Cloud security offerings: security assessment. secure Cloud architecture design, Authentication in cloud computing, Client access in cloud. Cloud contracting Model, Security Mapping, Identity Management, Case Study on Open Source & Commercial Clouds: Eucalyptus, Microsoft Azure, Amazon

M. C.A. 2nd Semester (2 Year Course) w.e.f.-2020-21
(Five Year Integrated Course)
Examination, May—2024
CLOUD COMPUTING

Paper- 20MCA22DB1

Time allowed : 3 hours] [Maximum marks : 80

Note Attempt five questions in all by selecting Question No. 1, which is compulsory and four questions from the rest by selecting at least one question from each unit. All questions carry equal marks.

1. (a) What is scalability? 80=16

(6) What is the concept of NFV?

(c) What is IDaaS?

(d) What is cloud cube model?

(e) What is significance of Platform as a Service? (D

What are the benefits of virtualization?

(g) What is Security Mapping?

(h) What is Utility Computing?

Unit-I

2. (a) What is cloud computing? What are its characteristics? What is cloud computing architecture? Illustrate. 12 (b) How did cloud computing evolve? Illustrate. 4

(4)

67062-N

3. Differentiate between the following:

(a) Grid Computing and Cluster Computing 8 (b)

Private and Public Clouds 8

Cait-II

4. (a) What are cloud services? What are its main types? Illustrate the significance of these services. 8

(b) What is virtualized environment? What are characteristics? Illustrate its significance. 8 5. Explain the following:

(a) Load balancing and performance Improvement 8

(b) Software Defined Networking (SDN) 8

Unit-III

6. (a) What do you mean by service level agreements (SLA)? What are SLA types? How are these managed? Illustrate their pros and cons. 8

(b) What are cloud services? What is cloud service lifecycle? What are main stages of life cycle? Illustrate. 8

7. Explain the following:

(a) Service Oriented Architecture (SOA) 8

(b) Network Management System features 8

67092-N

Unit-IV

- (a) What is cloud security? What are cloud security challenges? What are these challenges addressed? Illustrate. 10
- (b) What is cloud contracting model? Illustrate its significance. 6 Explain the following:
 - (a) Eucalyptus 8
 - (b) Amazon EC2 8

67062-

N

MCA 2nd Semester 2 Year Course
(w.e.f. 2020-21) Examination, May-2023

CLOUD COMPUTING paper-20MCA22DB1

Time

Time
N

allowed : 3 hours] [Maximum marks : 80

Note: Attempt five questions in all. Question No. 1 is compulsory and select four more questions from the rest, selecting one question from each Unit. All questions carry equal marks.

1. (a) What is Eucalyptus?
- (b) What is SON?
- (c) What are Cloudlets?
- (d) What is the concept of scalability?
- (e) What is Amazon EC2?
- (f) What is grid computing?
- (g) What is significance of CaaS?
- (h) What is NFV? State its significance. 80—16

Unit-I

2. (a) What is cloud computing? What are uses and applications of cloud computing? Illustrate through suitable examples. 8
- (b) What is cloud computing architecture? What are its main components? Illustrate. 8

67062-N-p-3-Q-9 (23)

(2) 67062-N

3. Explain the following Unit-IV

- (a) Cloud cube model 68. (a) What are security issues in Cloud Computing?
- (b) Cloud deployment models 10 are threats in a cloud computing environment? Illustrate. 8 Unit-II

4. (a) What is virtualization? What are its help? Illustrate. 8

- (a) Cloud Contracting
- (b) What is
- (b) Cloud Security Models 8

5. Explain the following:

- (a) Software Defined Networking (SDN) - 8
- (b) Load balancing 4
- (c) AMI 4

Unit-m

6. What are service level agreements(SLA)? What these differ? How are these managed? Illustrate Explain the following:

- (a) Large Scale Data Processing 8
- (b) Cloud services lifecycle management 8

(3) 67062-N

What

(b) What are different cloud security approaches? objectives? How do these

What are its benefits? Illustrate its significance.

9. Illustrate the following services provided by cloud: 10

Model 8

Hypervisor? What are its types? Discuss their relevance. 6

are SLA types? How do their pros and cons. 16

67062-N

67062-N

questions carry equal marks.

Compulsory Question

1. (a) What are cloud security challenges ?
(b) What do you mean by SOA ?
(c) What are web services ?
(d) What is Cloud-based big data analytics ?

- (e) What do you understand by security management in Cloud ?
- (f) What is the concept of load balancing ?
- (g) What is Amazon EC2 ?
- (h) What do you understand by security management in Cloud ? 2×8=16

Unit-I

2. (a) What is Cloud Computing ? How is it relevant in the context of Social Networks ? Illustrate.

Ddletcnuate Ptnate and i'ubbc
Clou&s8.8 the following qa' (-nuget

3. Compu•ar.g and relevance
&) Cäoud An•htteaun•

Unit-II

4. What ts Vituahzation What are Its
What are various types of Virtualization D;scuss ther relevance
5. Whar you mean by Cloud Scmces How
cloud compunng set•occs models are significant
What is the relevance of these services models

Unit-III

6. (a) What do you understand by Service Level Agreement (SLA) ? What is its significance ? Illustrate.
- (b) What do you mean by scalability ? What is its relevance in cloud services ? How is it achieved ? Illustrate. 8.8
7. How do you compare AWS, Azure and Google cloud platform ? Illustrate each of these highlighting their suitability with justification. 16

Unit-IV

8. What is cloud security ? What are the major security priorities in Cloud Computing ? What threats may be caused in a cloud computing environment ? How are these addressed ? Illustrate. 16
9. Explain the following :
- (a) Eucalyptus Cloud Platform
- (b) Design of Secure Cloud Architecture 8.2

paper : 20MCA22DB1

: 17 tree hours

Maximum Marks 80

For the questions, candidates should ensure they have supplied the correct and complete question NO in this regard, will be entertained after examination.

Mode : Attempt five questions in all. Question Number is compulsory. In addition to Compulsory question, student has to attempt four more questions selecting ~~one~~ question from each Unit.

I. (i) Distinguish between Internet and Cloud Computing



Virtualization ?

from Internet and Internet.

tagcous to the

(v) What is resource pooling and rapid elasticity?

(vi) What is the selection criterion for private cloud deployment model?

(vii) Write any two characteristics of public cloud and hybrid cloud.

(viii) Enlist the services that are provided by IBM Smart Cloud. 8×2

UNIT - I

2. (i) Define cloud computing. Enlist the characteristics and benefits of cloud computing. 8

(ii) Explain cloud computing with suitable diagram as well as explain its various applications in different areas. 8

3. (i) Explain different cloud deployment models with the help of diagram and enlist their benefits and limitations while implementing an application. 8

(ii) What are the advantages of "Software as a Service" (SaaS)? Explain with example. 8

UNIT – II

(i) Explain the different types of hardware virtualization techniques & their benefits. 8

- (ii) In context Of performance enhancement, explain the concept of load balancing in cloud computing. 8
- (i) Explain the architecture of Amazon web service (AWS) with the help Of diagram. Enlist its various applications in different fields. 8
- (if) Outline the various Objectives, benefits and limitation of virtualization. 8

UNIT - 111

(i) MCA 2nd sem
Cloud computing
management platform reference architecture and
8

(ii) What is the criterion for private cloud deployment model? 8

July 2022
Modu

- (i) Discuss service oriented prchitecture (SOA) with its advantages and disadvantages. 8
- (ii) Discuss about different databases and data stores in cloud. Also discuss SLA with its benefits and limitations. 8



ଉତ୍କଳ ବିଶ୍ୱବିଦ୍ୟାଳୟ
ଉତ୍କଳ ବିଶ୍ୱବିଦ୍ୟାଳୟ

8. (i) Discuss, in detail, the cloud security reference model and explain how cloud security is integrated into the design of application. 8
- (ii) Discuss the principles of security in cloud computing. 8
9. Explain the following : 16
- (i) Digital Signatures
- (ii) Goggle Cloud Platform

CLOUD COMPUTING

Paper : PE-CS-A402A / PE-IT-A404A / C

Subscribe



Time Allowed : 3 Hours

Maximum Marks : 75

Note : Attempt five questions in all. selecting at least one question from each Unit. All questions carry equal marks.

UNIT-I

Discuss various characteristics Of Cloud computing.
Write Cloud computing. Btech_cse notes_bank 7

- (b) Differentiate utility and distributed computing with examples.
- 2. (a) Discuss on "History of Cloud computing". 7
- (b) Differentiate grid and Cloud computing with examples.

UNIT-II

- 3. (a) Discuss various types of Protocols used in Cloud computing.
- (b) Differentiate Data Security and Network Security. S

48159/10967/650

P.T.n.

differentiate Community and Hybrid suitable examples.

- (b) Discuss various services offered by Cloud computing.

UNIT-

subscribe

involved in SLA. Discuss

5. (a) List different LA in Cloud computing. 8
components in detail the role of Eucalyptus. 7
- (b) Discuss Case study on ERP with examples. 7
6. (a) Differentiate CRM and ERP in Cloud computing ?
Database Btech cse notes 8
- (b) How the data is stored in Cloud computing ?
Discuss various techniques in Cloud. 7

UNIT- Level and Application Level

7. (a) Discuss Cloud contracting detail. 8
- (b) Differentiate Network Security. 8
8. (a) Explain IAM model in detail. 7

Discuss various location.

differentiate Community and Hybrid suitable examples.

- (b) Discuss various services offered by Cloud computing.

UNIT-

subscribe

involved in SLA. Discuss

5. (a) List different LA in Cloud computing. 8
components in detail the Eucalyptus. 7
role of
- (b) Discuss Case study on ERP with examples. 7
6. (a) Differentiate CRM and ERP in Cloud computing ?
Database Btech cse notes 8
- (b) How the data is stored in Cloud ?
Discuss various techniques in Cloud. 7
model in detail,

UNIT-Level and Application Level

7. (a) Discuss Cloud contracting detail. 8
8
- (b) Differentiate Network Security. 7
functional issues raised in Data
8. (a) Explain IAM model in detail.

Discuss various

location.

Unit-I

67062-N

2. (a) What is NIST model of cloud computing? Illustrate relevance of its each element. 8
- (b) How Private Clouds are different from Public Clouds and Hybrid Clouds? Illustrate their usefulness. 8
3. Explain the following: 8
- (a) Cluster Computing and its relevance 8
- (b) Cloud Architecture 8

Unit-II

4. (a) What is virtualization? What are various types of virtualization? Discuss their relevance. 8
- (b) What is the concept of Software Defined Network (SDN)? How is it significant in cloud environment? Illustrate. 8
5. Explain the following:
- (a) Platform as a Service and its benefits 8
- (b) Load Balancing 8

67062-N

MCA 2nd Semester 2 Yr Course (w.e.f.) 2020-21

(Fresh & Reappear)Examination, May-2026

Cloud Computing

Paper : 20MCA22DB1

Time allowed : 3 Hours] [Maximum marks : 80

Note: Question No. 1 is compulsory. Attempt four questions by selecting one question from each Unit. All questions carry equal marks.

1. (a) What is Grid computing? 8×2=16
- (b) What do you mean by IDaaS?
- (c) What are Hypervisors?
- (d) What is Cloud-based data analytics?
- (e) What do you mean by security management in Cloud?
- (f) What is Cloud contracting model?
- (g) What do you mean by Cloud Service Catalog?
- (h) What are web services?

67062-N-P-3-Q-9 (26)

[P.T.O.]

Unit-III

6. (a) What is Service Level Agreement (SLA)?
What is its significance? Illustrate. 8
- (b) What do you mean by scalability? What is
its relevance in cloud services? How is it
achieved? Illustrate. 8
7. (a) What is cloud management? What are
different stages of lifecycle management of
cloud services? Illustrate their relevance. 8
- (b) What is Service Oriented Architecture
(SOA)? What role does it play in cloud
service management? Illustrate. 8

Unit-IV

8. (a) What is cloud security? What are the major
security priorities in Cloud Computing?
Illustrate. 8
- (b) What are the main cloud security
challenges? What security techniques are
followed to address these challenges?
Illustrate. 8
9. Explain the following: 8
- (a) Amazon EC2 and its significance 8
- (b) Eucalyptus Cloud Platform